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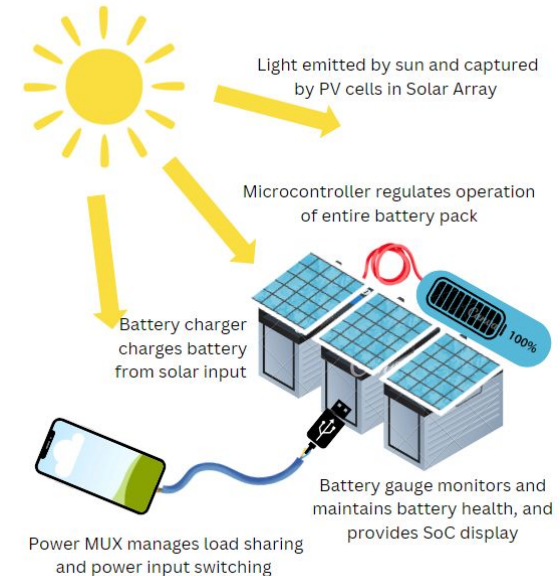
ENGINEERING
TEXAS A&M UNIVERSITY

Team 64: TI Solar Battery Pack Bi-Weekly Update 1

**Jaein Cha, Patrick Wang, Cole Miller, Camden Pomeroy
Sponsor: Wyatt Keller**

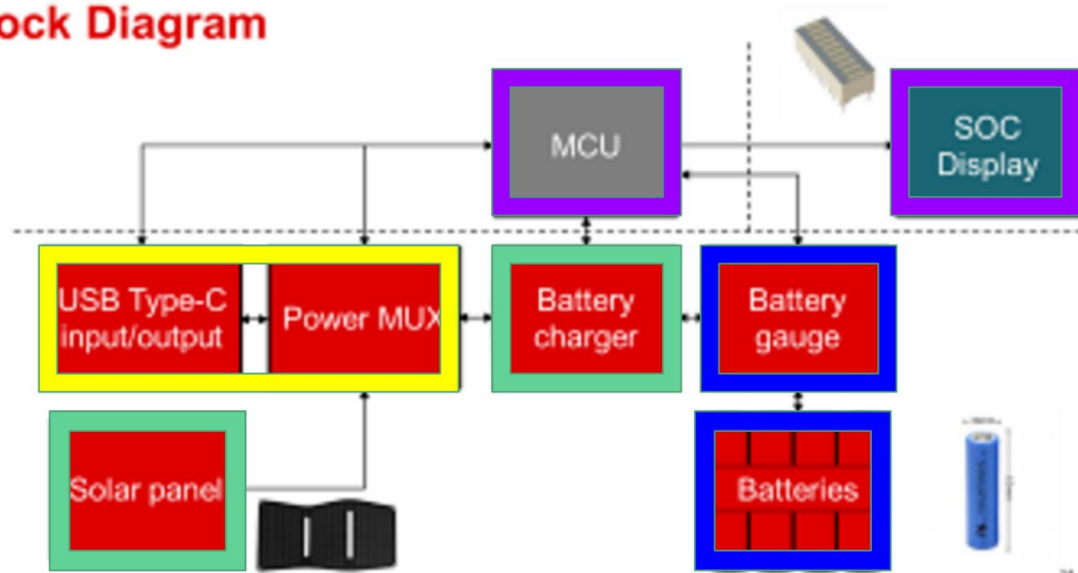
Project Summary

- Natural disasters can cause unexpected power outages resulting in the inability to charge cell phones and other necessary small electronic devices.
- A solution is to create a solar powered battery pack that charges itself from solar panels, and distributes the power to electronic devices through a USB connection.



Project/Subsystem Overview

Block Diagram



Camden ■
Cole ■
Patrick ■
Jaein ■



Project Timeline



January

Individual Subsystem Finalization

Wrap up subsystem design and
validation testing



February

Integration Design

Work together to integrate the
designs of each subsystem



March

Assembly/Validation

Assemble the final pcb and 3D
printed enclosure. Perform
validation testing and identify
problems that occur



April

Validation/Debugging

Engineer a solution to problems
that arise as a result of
integration



Battery Gauge / Battery Pack

Patrick Wang

Accomplishments since 403 15 hrs	Ongoing progress/problems and plans until the next presentation
• Ordered new 1s3p battery pack for the batteries • Ordered new pcb with more pins for battery pack connections	• Turned FETs on, board sparked and I2C connection stopped • Solder new board with IC and new parts and finish up last validation test

Battery Gauge / Battery Pack

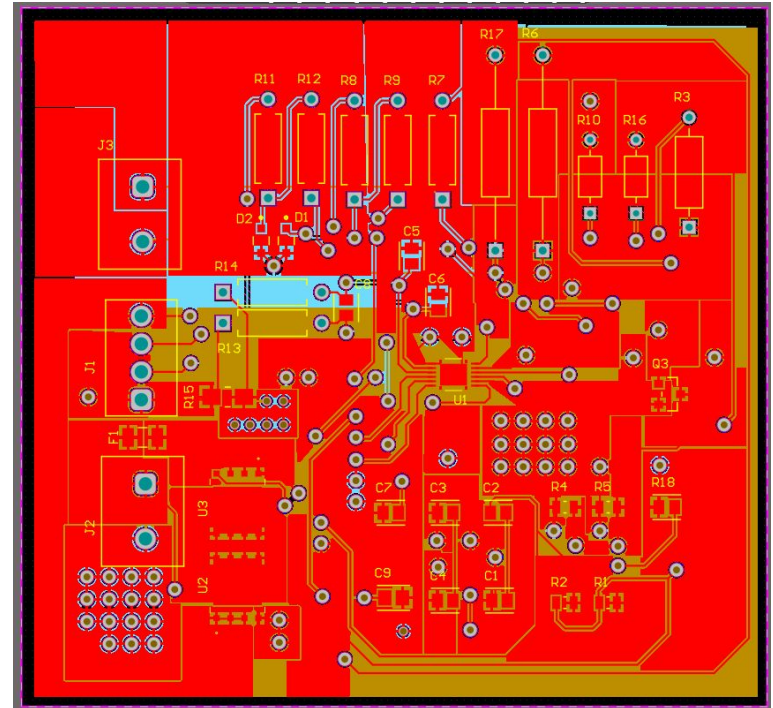
Patrick Wang

What works:

- Board allows in-threshold input and output voltage through to charge battery pack
- Current throughput also correct

What doesn't work:

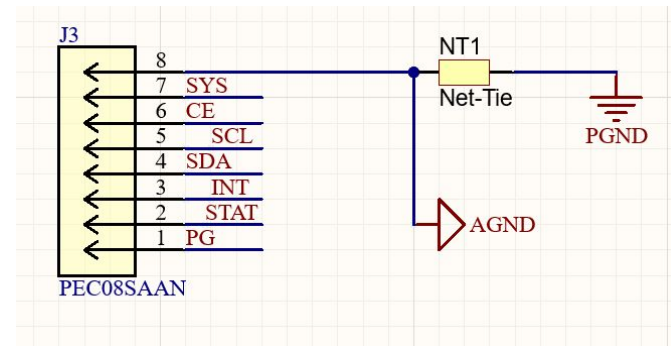
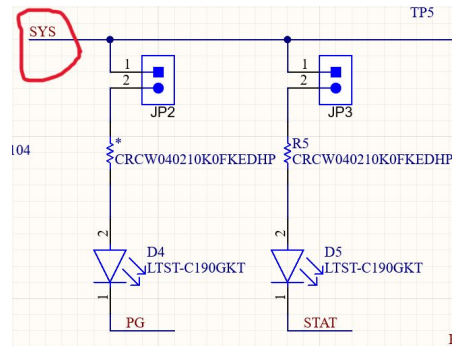
- Board stopped communicating with I2C interface, may be something wrong with SCL and SDA pins
- Will solder new board to fix issue



Battery Charger/Solar Panel

Cole Miller

Accomplishments since 403 15 hrs of effort	Ongoing progress/problems and plans until the next presentation
– Troubleshoot digital pins – Added Reset function and net tie – Ordered new board – Hardwired board to rely on own voltage	– Assemble new board – Finish validation – Update net names



Battery Charger/Solar Panel

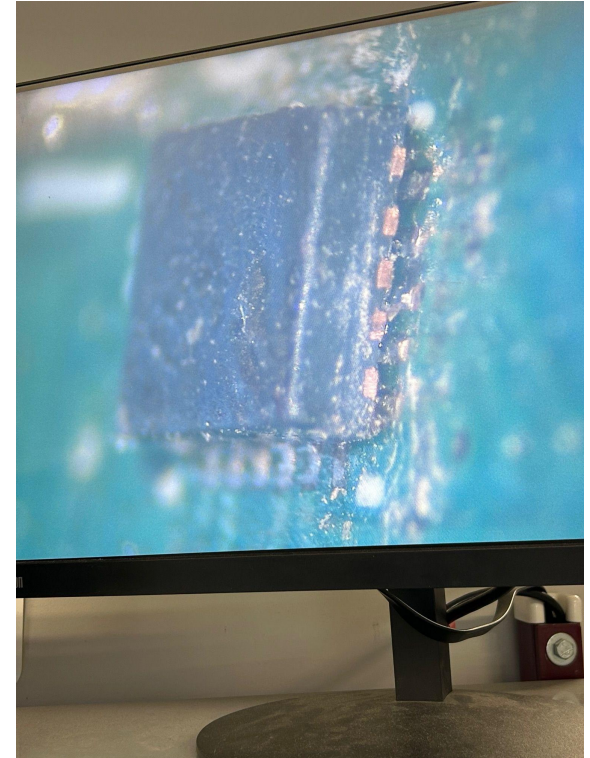
Cole Miller

What works:

- Buck converter successfully switches voltages
- Validated all other pins as operational
- Solar panel works and meets specs

What doesn't work:

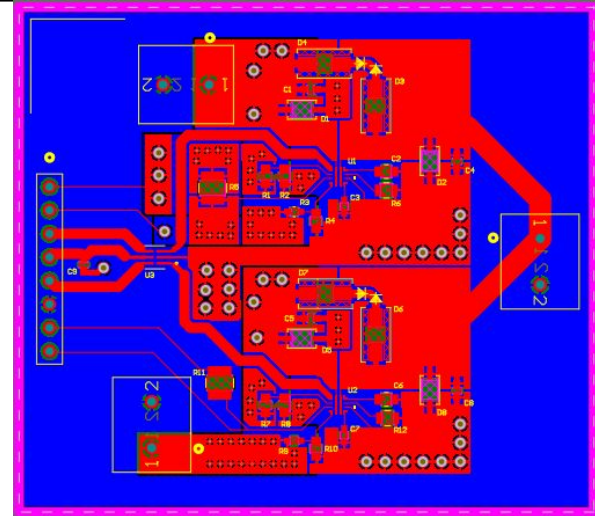
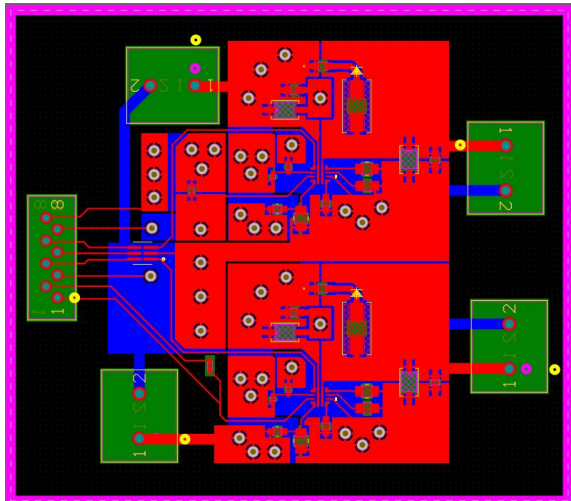
- Cannot operate digital Pins
- Cannot communicate with EVM
 - Weak connection as shown



Power MUX

Camden Pomeroy

Accomplishments since 403 15 hrs of effort	Ongoing progress/problems and plans until the next presentation
Redesigned PCB layout Ordered new PCB Ordered new components	Assemble/Validate new PCB Work towards initial integration



Power MUX

Camden Pomeroy

What works:

- Input Voltage = Output Voltage
- UVLO/OVLO Operates as expected
- EN pin opens and shuts off properly

What doesn't work:

- TMUX not fully validated
- Two VBUS paths
- Could not draw current on previous board

Number	VALIDATION PLAN	Input:	Excpeted Output:	Results	Status
1	PP1 Operational	4-6V 2.5-3.5A on Vin (PP1)	4-6V 2.5-3.5A on Vout (PP1)	4.5V of Vout when input was 5V	
2	PP2 Operational	4-6V 2.5-3.5A in Vin (PP2)	4-6V 2.5-3.5A on Vout (PP2)	As expected	
3	PP1 PG (Good signal)	If Vin > 2V	PG signal is Low	As expected	
4	PP1 PG (Bad signal)	if Vin < 2V	PG signal is High	As expected	
5	PP2 PG (Good signal)	If Vin > 2V	PG signal is High	As expected	
6	PP2 PG (Bad signal)	if Vin < 2V	PG signal is Low	As expected	
7	PP1 UVLO Operational	if Vin < UVLO	No output	As expected	
8	PP2 UVLO Operational	if Vin < UVLO	No output	As expected	
9	PP1 OVLO Operational	if Vin > OVLO	No output	As expected	
10	PP2 OVLO Operational	if Vin > OVLO	No output	As expected	
11	SEL PP1	SEL is high	PP1 is active and PP2 is not	Sel pin would not properly switch between S1 and S2	
12	SEL PP2	SEL is low	PP2 is active and PP1 is not	Sel pin would not properly switch between S1 and S2	
13	ILM during startup	Inrush current of 4A	Limited current (<4 A) to ram	Could not get DC load to operate	
14	Seperate Power Paths	S1 active and both Vins 5V	Only PP1 Vout = 5V	As expected	
15	Seperate Power Paths	S2 active and both Vins 5V	Only PP2 Vout = 5V	As expected	
16	PGTH Threshold	Test Voltage of PGTH	PGTH 3.5V	As expected	



Microcontroller Unit

Jaein Cha

Accomplishments since 403 15 hrs of effort	Ongoing progress/problems and plans until the next presentation
– Verified problems with the PCB – Resoldered MCU – Updated PCB design – Verification of updated design with the sponsor	– Order updated PCB & parts – New PCB assembly – Validate the new PCB – Verify I2C functionality with other subsystem boards

Microcontroller Unit

Jaein Cha

What works:

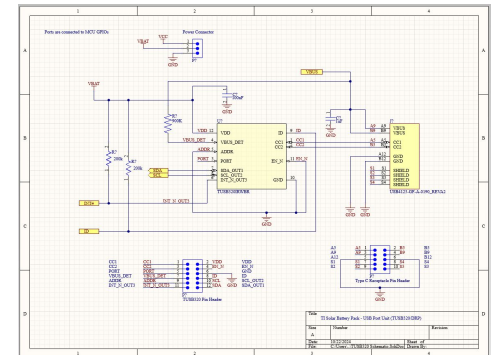
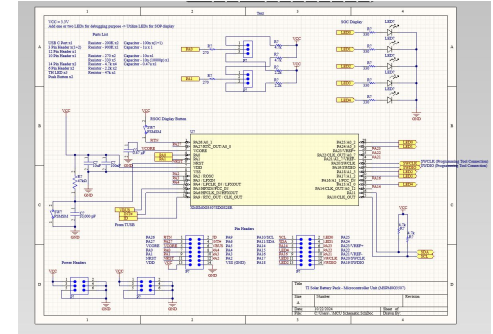
- I2C firmware between the MCU and the battery gauge EVM
- Input (button) and output (LEDs) on the board when connected to the launchpad

What doesn't work:

- MCU is not connected to the pads properly - not responsive

What is changing:

- Adjustment on the power trace (VBUS net) - width, add connector
- Size of the board - reducing the size





Parts Ordering Status

- Ordered new PCBs for
 - Battery Charger
 - Battery Gauge
 - Power MUX
- Ordered new 1s3p battery pack holder
- Ordered various assembly components for each subsystem

Execution & Plan

[illegible]



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Questions?